

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1069

Folsom Prison 138 kV
October 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1069. The Interconnection Customer proposes a 83.8 MW solar photovoltaic facility interconnecting at the Folsom Prison 138 kV substation in the West Virginia area. The request was received on September 28, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 277 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,034,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

DECLO to NEW LONDON 138kV: 116.2% loading (425 MVA) following the outage of the BATH COUNTY to SHERBURNE 345kV; DFAX = 14.7%.

For the loss of ASPEN SUBSTATION to LOWMOOR 345kV, the AUSTIN to BATH COUNTY 138kV Cir. 3 loads to 104.6% of its 311 MVA rating. Project DFAX: 8.3%.

For the loss of ASPEN to LOMOOR 345kV, the BATH COUNTY to AUSTIN 138kV Circuit 1 loads to 102.9% of its 316 MVA rating. Project DFAX: 8.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DECLO to NEW LONDON 138kV	Rebuild line with higher-capacity conductor	\$40,331,000	4% (\$1,775,000)
AUSTIN to BATH COUNTY 138kV Cir. 3	Reconductor line	\$5,618,000	27% (\$1,498,000)
BATH COUNTY to AUSTIN 138kV Circuit 1	Reconductor line	\$2,854,000	27% (\$761,000)
	Total allocated to project		\$4,034,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.